Design Report for Silicon Photonics Edx Course

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***Abstract*—As part of the UBCx Edx course, "Silicon Photonics Design, Fabrication and Data Analysis", the students are designing, manufacturing and analyzing a Mach-Zehnder interferometer. This document presents the design proposal of one of the students.**

***Index Terms*—Silicon photonics, Mach-Zehnder interferometer**

I. INTRODUCTION

As part of the UBCx Edx course "Silicon Photonics Design, Fabrication and Data Analysis" the following design is presented of a Mach Zehnder interferometer. The goal of the design is to fabricate and measure several types of MZI circuits each with changed length difference and observe the change in free spectral range. The first MZI type uses y-splitter at one side and directional couple on the other side, creating two complementary outputs. The second MZI type uses y-splitter at both it's input and output. Loop back circuit and y branch only circuit are also implemented for calibration.

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II. THEORY

As explained in [1],[2] and repeated for completeness in the Appendix, the interferometer transfer function for both MZI described above is

$$\frac{I_o}{I_i} = \frac{1}{2}(1 + \cos(\beta \Delta L)) \tag{1}$$

The free space spectral range (FSR) which is the spacing between adjacent peaks can be calculated to be

$$FSR = \frac{\lambda^2}{\Delta L n_g} \tag{2}$$

Thus, by measuring the MZI FSR it is possible to extract the waveguide group index.

The group index can be found by forming compact model for the waveguide effective index dispassion. Using Lumerical Interconnect a frequency sweep simulation can be performed and the resulted effective index as function of wavelength can be exported to MATLAB. In the course [2], a MATLAB script is provided to find the Taylor expansion polynomial

$$n_{eff}(\lambda) \approx n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \tag{3}$$

The Taylor coefficients can be related to the physical parameters effective index, group index, and dispersion as described in the course notes [2]

$$\begin{aligned} n_{eff} &= n_1 \\ n_g &= n_1 - n_2\lambda_0 \\ D &= -\frac{2\lambda_0 n_3}{c} \end{aligned} \tag{4}$$

The design files were submitted as part of the SiEPIC openEBL-2026-05 fabrication review process using the public GitHub repository.

III. MODELING AND SIMULATIONS

A. Waveguide Model

The MZI is implemented using strip waveguide, TE polarized with width of 0.5um and height of 0.22um. The first TE mode profile and dispersion were calculated using Lumerical mode and are plotted next

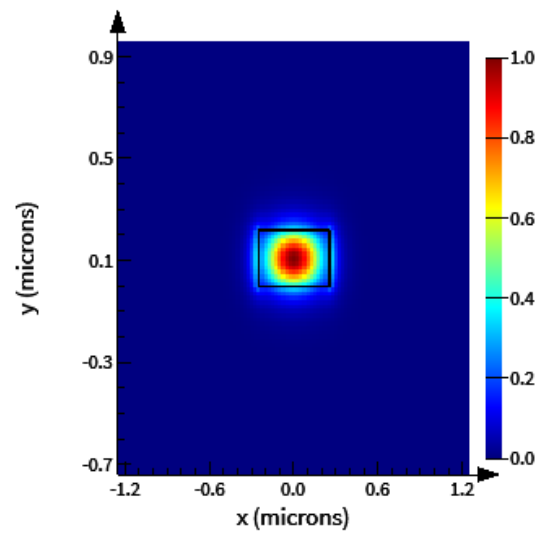

Fig. 1. Mode Profile.

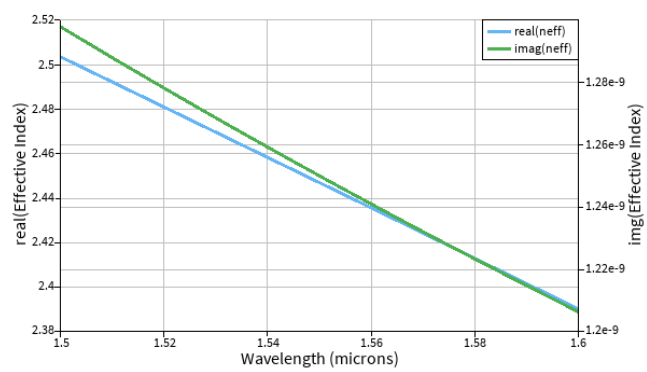

Fig. 2. Effective index versus wavelength.

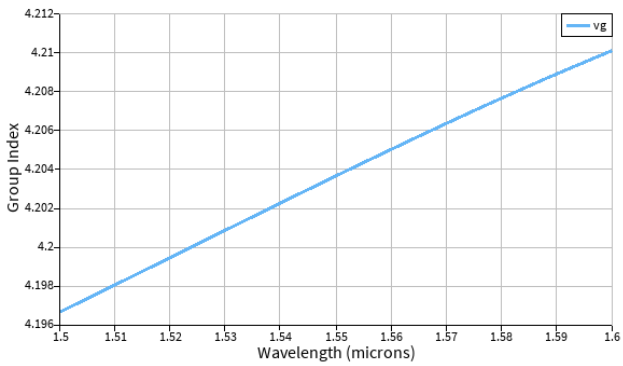

Fig. 3. Group index versus wavelength

The effective was exported to MATLAB. A MATLAB script provided by the course was used to fit a compact model for the waveguide effective index with the following result

$$n_{eff}(\lambda) \approx 2.45 - 1.13(\lambda - 1.55) - 0.049(\lambda - 1.55)^2 \quad (5)$$

From the above compact model and using (4) the group index can be calculated to be

$$n_g = 2.45 + 1.13 * 1.55 = 4.2$$

Which is similar to the group index simulated with mode as shown in figure (3).

B. Layout and MZI design variations

There are total of seven MZIs implement in the layout. MZI1 and MZI2 are implemented using y-splitter at the input and wideband directional coupler at the output. The path difference is 100um and 150um. MZI3 to MZI7 are implemented with y-splitter at both input and output and path differences of 50um, 75um, 100um, 150um and 200um. In addition, a loop back is implemented to measure the grading coupler response and an circuit to measure the response of the y splitter used.

TABLE I
LAYOUT VARIATIONS

MZI #	path difference [um]	FSR [nm]	MZI Type
1	100	5.7	Y-splitter plus coupler
2	150	3.8	Y-splitter plus coupler
3	50	11.4	y-splitter
4	75	7.6	y-splitter
5	100	5.7	y-splitter
6	150	3.8	y-splitter
7	200	2.9	y-splitter

Below is the layout as plotted in Klayout

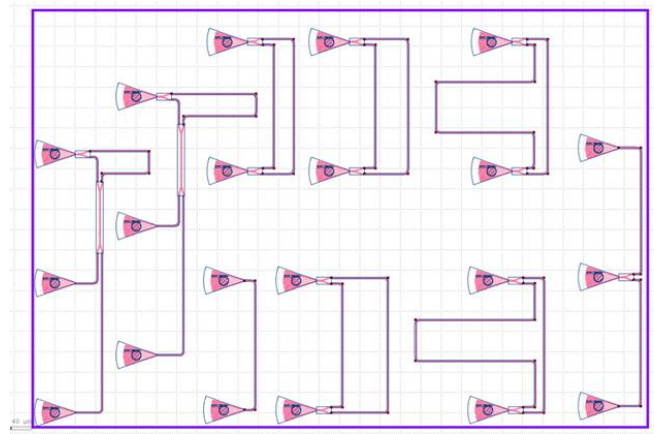

Fig. 4. Layout

C. Transmission spectrum simulation

Transmission spectrum of MZI5, which have path difference of 100um was simulated with Lumerical interconnect with below results

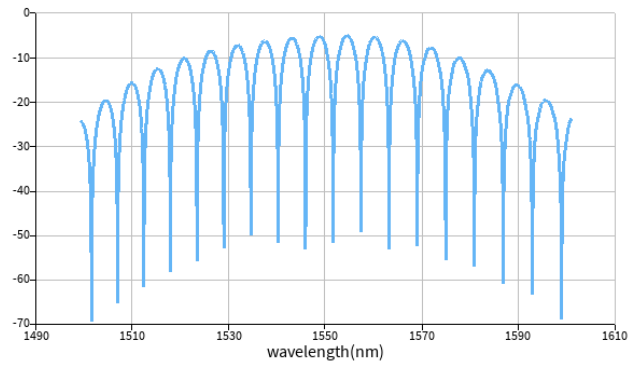

Fig. 5. MZI5 simulated transmission spectrum

From the simulated spectrum the FSR is seen to be 5.8nm, similar to the expected calculated value of 5.7nm.

IV. FABRICATION

to be completed later

V. EXPERIMENT DATA

to be completed later

VI. ANALYSIS

to be completed later

VII. CONCLUSION

to be completed later

APPENDIX

The important formulas derivation used in this report are summarized here, they are reproduction of information available at [1] and [2].

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A. MZI Transfer Function

In a MZI topology the input light is split using a y-spitter, then each half of the light pulse propagates over a different length waveguide, the two waveguides are then combined using a second y-combiner yielding the output. For input intensity and field of Ii and E_i and neglecting loss the fields at the two inputs of the second combiner arms after each propagated over different length are

$$E_{01} = \frac{E_i}{\sqrt{2}} e^{-i\beta L}$$

And

$$E_{02} = \frac{E_i}{\sqrt{2}} e^{-i\beta(L+\Delta L)}$$

Where β is the propagation constant related to the index of refraction n with

$$\beta = \frac{2\pi n}{\lambda}$$

When both fields are combined, the output intensity becomes

$$I_o = \frac{I_i}{4}\left|e^{-i\beta L} + e^{-i\beta(L+\Delta L)}\right|^2 = \frac{I_i}{2}(1 + \cos(\beta\Delta L))$$

The distance between the peaks of this function is called the free spectral range.

B. The Free Spectral Range of Imbalanced MZI

Assuming two adjacent peaks of the MZI transfer function are located at wavelengths λ_{m+1} and λ_m where the distance between them is defined as the FSR

$$FSR = \Delta\lambda = \lambda_{m+1} - \lambda_m$$

Assuming the propagation constant for each of the above adjacent wavelength are β_{m+1} and β_m than the different between the two propagation constants is defined as

$$\Delta\beta = \beta_m - \beta_{m+1}$$

From the MZI transfer function the phase difference between adjacent peak must be 2π requiring

$$\Delta\beta\Delta L = 2\pi$$

Thus

$$\Delta\beta = \frac{2\pi}{\Delta L}$$

Assuming the distance between the MZI arms is much smaller than the difference between the arms we can approximate the relation between Δβ and Δλ using a derivative

$$\frac{\Delta\beta}{\Delta\lambda} \approx -\frac{d\beta}{d\lambda} = -\frac{d}{d\lambda}\left(\frac{2\pi n}{\lambda}\right) = \frac{2\pi}{\lambda^2}\left(n - \frac{dn}{d\lambda}\lambda\right) = \frac{2\pi}{\lambda^2} n_g$$

Where n_g is the group index.

Thus we can calculate the FSR as follows

$$\Delta\lambda = \frac{\Delta\beta}{\frac{2\pi}{\lambda^2} n_g} = \frac{\frac{2\pi}{\Delta L}}{\frac{2\pi}{\lambda^2} n_g} = \frac{\lambda^2}{n_g \Delta L}$$

References

[2] Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015

[3] L. Chrostowski, "Silicon Photonics Design, Fabrication and Data Analysis," UBCx, edX. Accessed: May 4, 2026. [Online]. Available: https://www.edx.org/learn/engineering/university-of-british-columbia-silicon-photonics-design-fabrication-and-data-ana